Setting up Stereoscopic Rendering with Unreal nDisplay for LAVA's 3D Display Wall (and Samsung Odyssey 3D display)

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Version 05/05/2026

This document provides a step-by-step guide for setting up stereoscopic rendering on LAVA's 3D display wall using Unreal Engine's nDisplay, covering both new projects and integrating nDisplay into existing ones. It explains how to configure nDisplay assets, Switchboard launching, and optional low-level setup details.

Before you begin, be aware that nDisplay has not been implemented for the Mac.

There are two ways to start:

1. Start with Part A to build a **new** Unreal 3D Display Wall project.
2. Start with Part B if you want to **add** LAVA's nDisplay capability or nDisplay support for the Samsung Odyssey 3D display, to an **existing** Unreal project.

Then continue with the following parts:

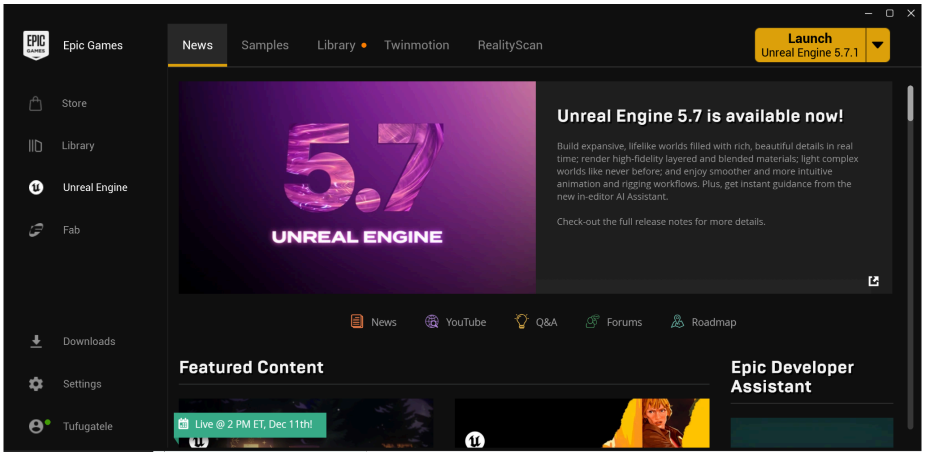
- Part C - Adding LAVA's nDisplay Asset or nDisplay asset for Samsung Odyssey 3D to Existing Unreal Project
- Part D - Set up the Switchboard and Switchboard Listener to launch nDisplay
- Part F - Tips and Caveats

If you wish to understand how the LAVA nDisplay asset is implemented, go through Part A, especially Part E, D and F.

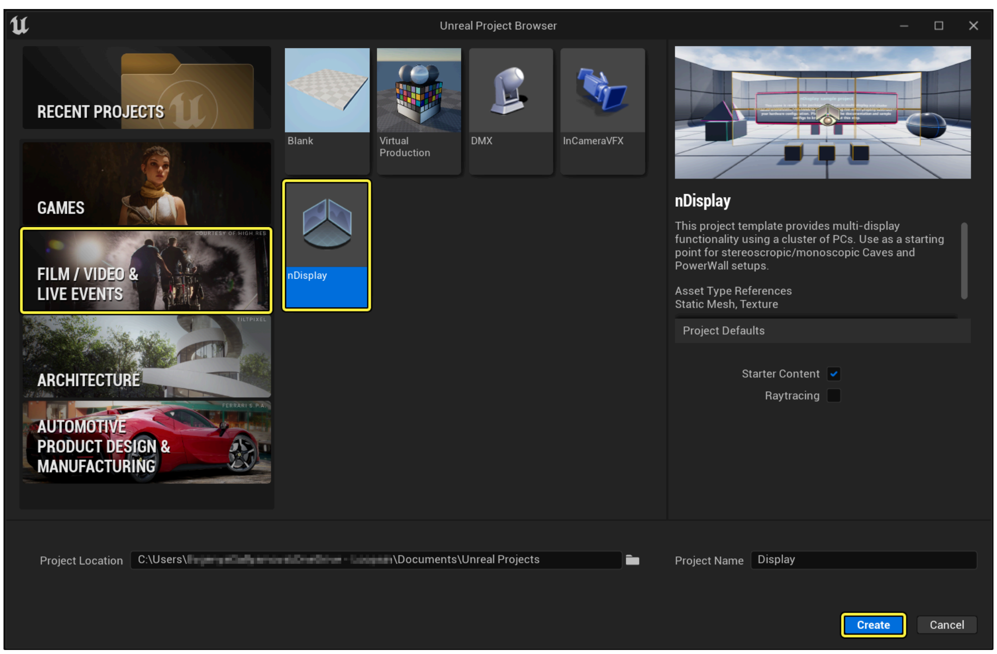
Part A. Getting Started with nDisplay from Unreal nDisplay Project Template

Unreal's nDisplay project template is essentially their simulation template with the nDisplay plugins already installed. To install the nDisplay project, follow these steps:

1. Go to Epic Games website and download their hub. From the hub install the latest version of Unreal.
2. Having installed Unreal, launch it by clicking on the Launch Unreal Engine button



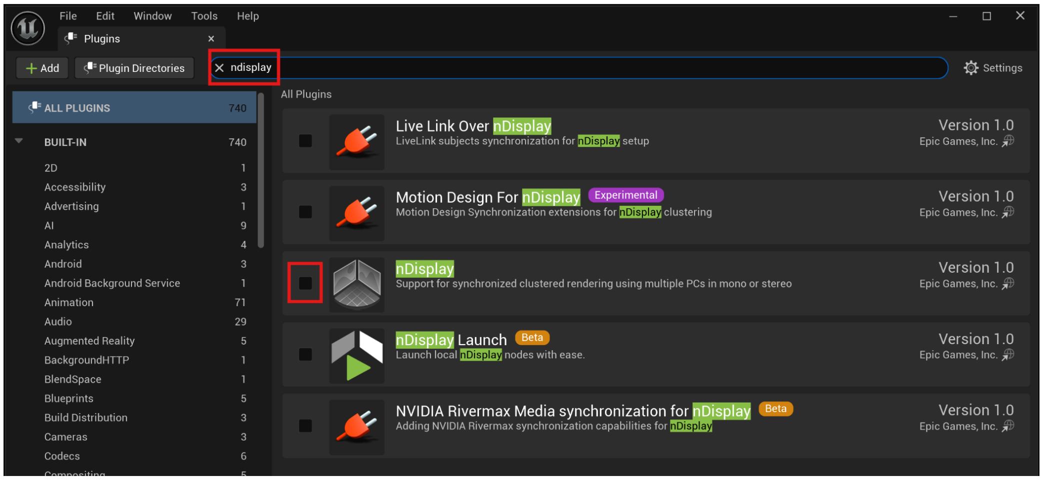
3. Create an nDisplay template project as follows: Click on **New Project**. Then click on **Film/Video & Live Events**. Choose the **nDisplay** template project and enter a suitable project name.



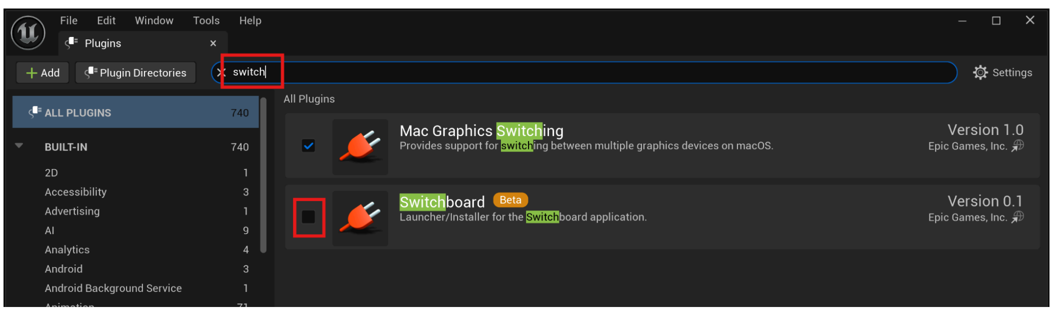
4. Once the project is launched, look for the NDC_Basic actor in the Outline tab of your project. If you intend to go onto Part C, you can delete the NDC_Basic actor. If you plan to go on to Part E do not delete NDC_Basic actor.

Part B. Adding nDisplay and Switchboard Plugins to an Existing Unreal Project

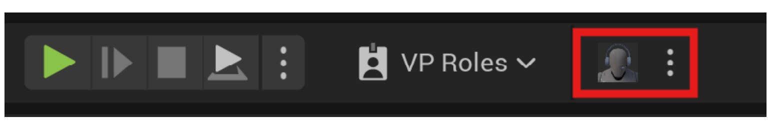
1. From your Unreal editor, go to the **Edit** menu and click **Plugins**.
2. Search for the **nDisplay plugin** and enable it. You may have noticed that there is an nDisplay Launch plugin too. Do not enable that. It's something you can explore later.



3. Now search for the **Switchboard plugin** and enable that too.



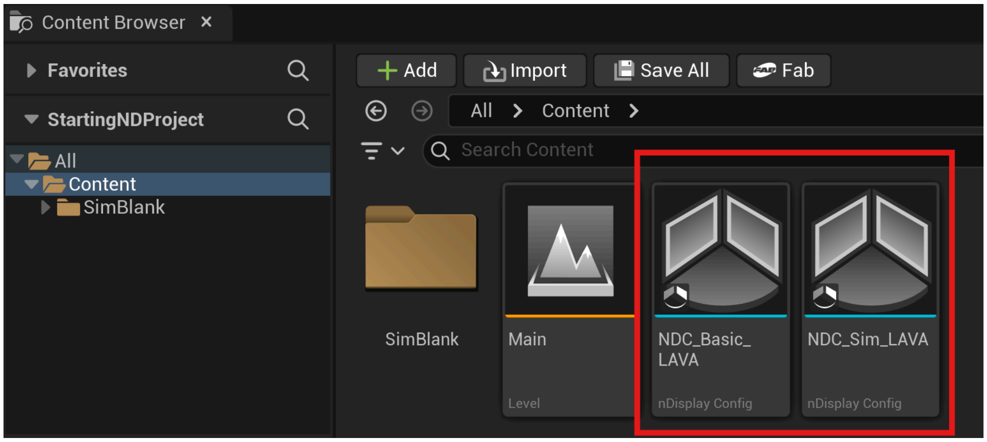
4. By now, you will likely also notice a prompt at the bottom of the plugin window asking you to restart Unreal. Go ahead and click on **Restart**.
5. You should see this at the top of your Unreal editor. Part D will explain how to use those buttons.



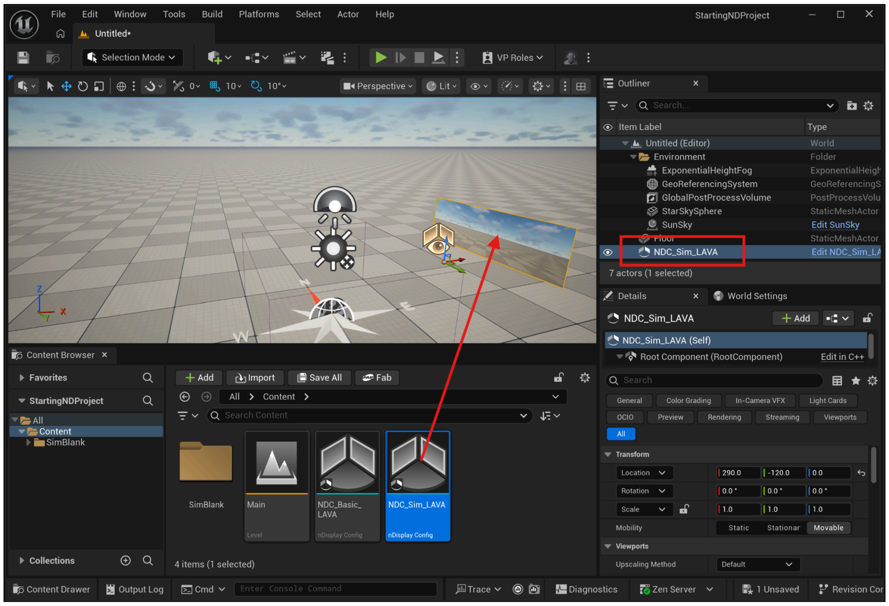
Marked in RED are the controls for the Switchboard- which will be explained in Part D.

Part C. Adding nDisplay Support and LAVA nDisplay Asset to Existing Unreal Project

1. Download the **NDC_Basic_LAVA.uasset, NDC_Sim_LAVA.uasset, and NDC_Samsung_LAVA.uasset** files. (https://github.com/doctorspiffy/Unreal-nDisplay-3D-Wall) Copy and paste them into the **Content** folder of your UnReal project. You'll see them appear in your Unreal editor's **content browser tab**.



2. Drag either NDC_Basic_LAVA, NDC_Sim_LAVA, NDC_Samsung_LAVA asset into the Unreal editor's **scene** window to turn it into an actor, and then position and orient the resulting nDisplay preview window as appropriate. Note the following difference between NDC_Basic_LAVA and NDC_Sim_LAVA:
   - The **NDC_Basic_LAVA** asset contains the nDisplay configuration to drive LAVA's display wall which requires you to render two side-by-side stereo pairs of 4800x1620 each- making a total resolution of 9600x1620. Such resolution will require substantial graphics memory and may choke a development laptop.
   - The **NDC_Sim_LAVA** asset contains an nDisplay configuration of a scaled down display wall that you can use to make development possible on computers with lesser graphics capability.
   - The **NDC_Samsung_LAVA** asset contains an nDisplay configuration for the Samsung Odyssey 3D display. If you have the Pro version of the Acer SpatialLabs display, this asset will likely also work for that display as well.



3. Now go to the Unreal Editor's **File** menu and click **Save All**.

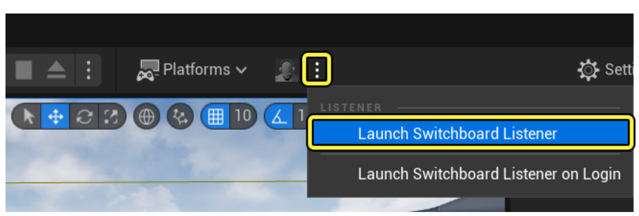
Part D. Set up the Switchboard and Switchboard Listener to launch nDisplay

If you attempt to run your program by clicking on Play in the Unreal Editor, it will run the program but not by using nDisplay. To run via nDisplay, you need to use the Switchboard.

The switchboard controls the devices (computers) that drive the displays in your VR-room set up (whether it's a single display wall or a CAVE). Switchboard does this by communicating with Switchboard Listeners, potentially running on separate computers, to launch copies of your project. For a single display wall, both the Switchboard and Switchboard Listener will reside on the same computer. For more information on nDisplay visit:
https://dev.epicgames.com/documentation/en-us/unreal-engine/ndisplay-overview-for-unreal-engine

The following are instructions for setting up Switchboard and the Listener:

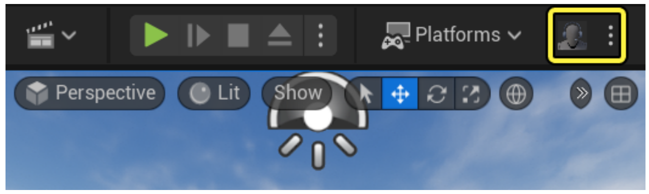
1. In the Unreal editor, go to the Toolbar, click the 3 dots next to Switchboard and choose **Launch Switchboard Listener**.



2. Switchboard Listener launches and may immediately minimize itself. You may also be asked to provide a password. If you are not asked for the password then your organization may already have set one, so ask your administrator. Actually, if you look at the Settings menu in the listener there is an option for you to change the password if

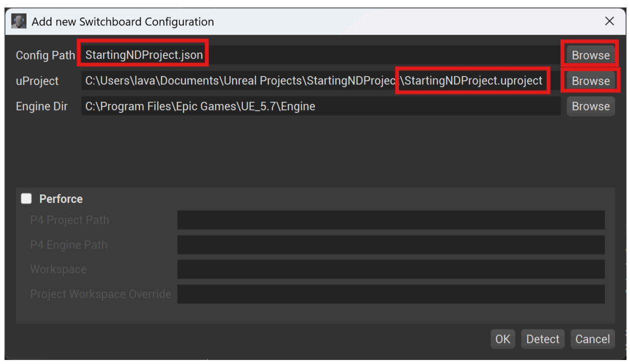
necessary. Later this password will be used to authorize the connection between the Switchboard and the Switchboard Listener.

3. Anyway, continuing- once the switchboard listener is open, minimize it (don‘t close it).
4. In the Unreal editor Toolbar, click the Switchboard button to launch the Switchboard application on your computer.

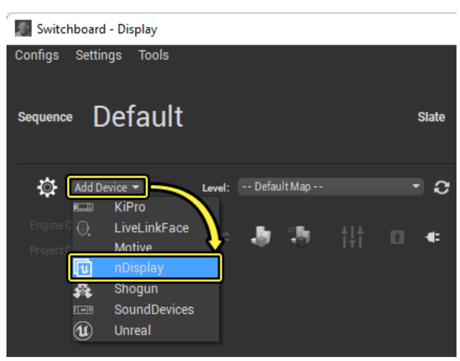


When launching Switchboard for the first time, a command prompt window may appear, installing required dependencies before the Switchboard window opens. Allow the installation to proceed.

5. When the switchboard opens, the **Add new Switchboard Configuration** window will likely appear. A switchboard **configuration** is needed for each Unreal project that will need to use nDisplay. If this window does not appear go to the Switchboard‘s **Configs** menu and click **New Config**.

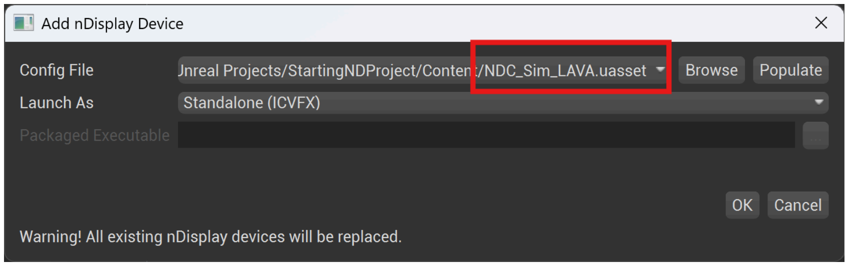


6. Give a name to **Config Path** field- by clicking on the Browse button next to the field.
7. Then in the **uProject** field, browse to your project‘s .uproject file.
8. Then click OK to proceed.
9. In Switchboard, click **Add Device** in the top left and choose **nDisplay** to open the Add nDisplay Device window.

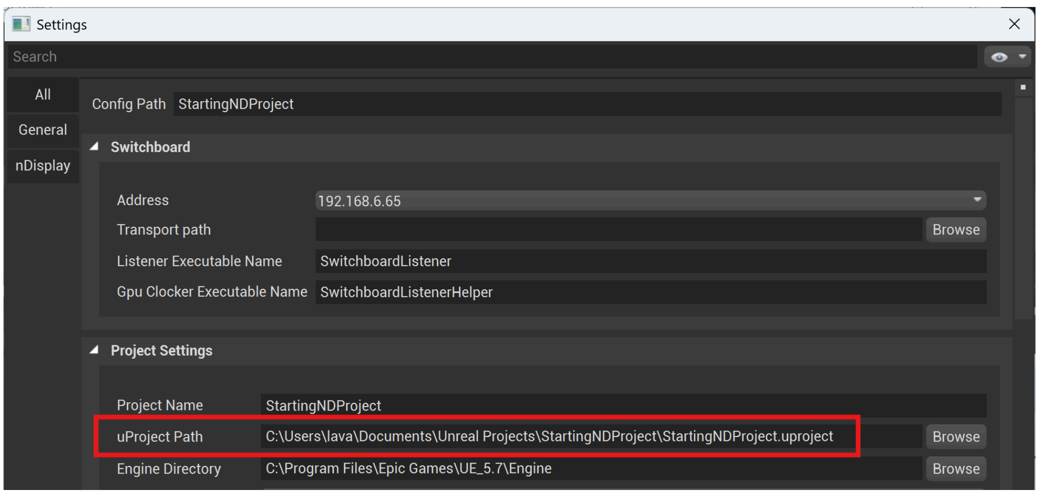


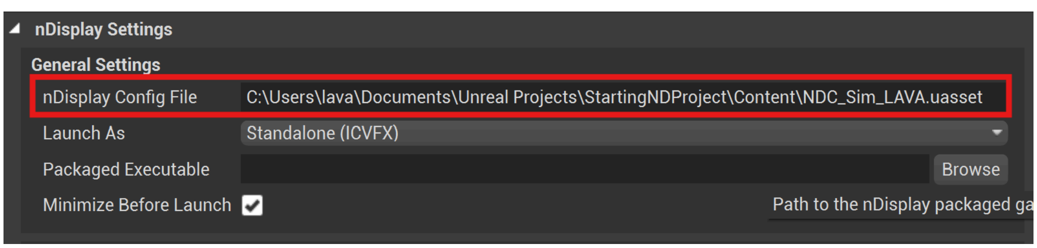
10. Browse for and select the nDisplay .uasset configuration file in your project (either **NDC_Basic_LAVA.uasset, NDC_Sim_LAVA.uasset, or NDC_Samsung_LAVA.uasset**), and click **OK**. If you are doing this for the first time, I

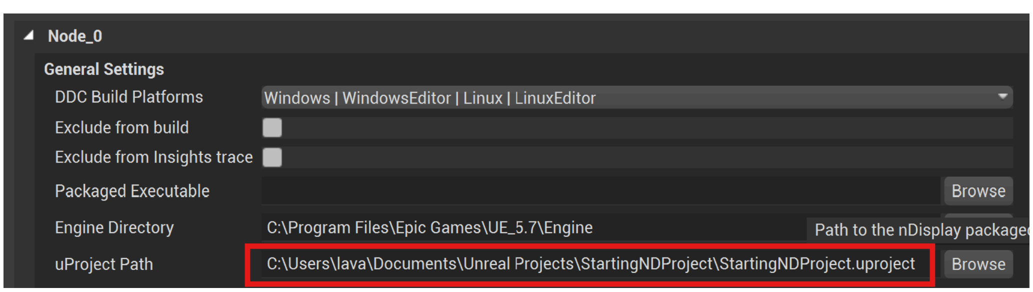
recommend using NDC_Sim_LAVA. NDC_Basic_LAVA is the configuration for the actual display wall. NDC_Sim_LAVA is intended as a lower resolution simulator that can run on weaker laptops/PCs. Later you will want to repeat the above steps in Part D to create a separate device configuration for NDC_Basic_LAVA or NDC_Samsung_LAVA.



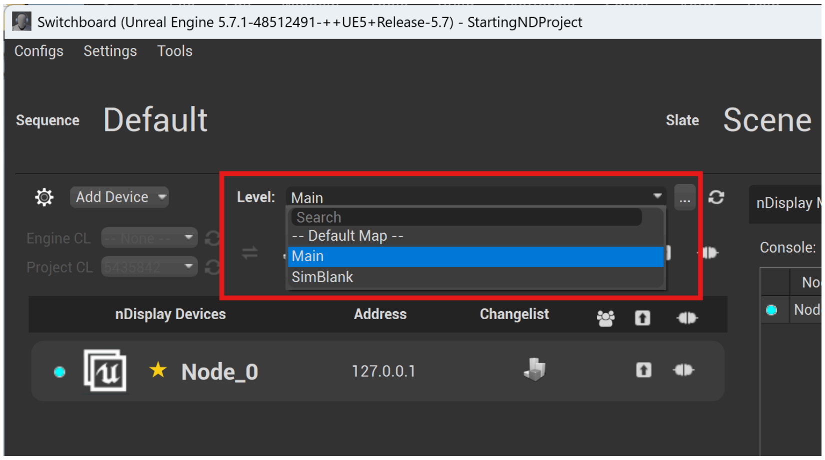
11. Continuing- now click on the **Settings** menu in the Switchboard and make sure the **Project Settings** sub-panel, the **nDisplay Settings** sub-panel, and the **Node_0** sub-panel contain the correct information. Below are examples of what you might see. Notice the paths show the location of your project file and the nDisplay uasset.



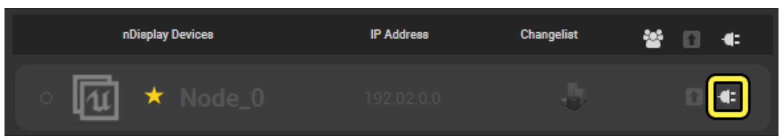




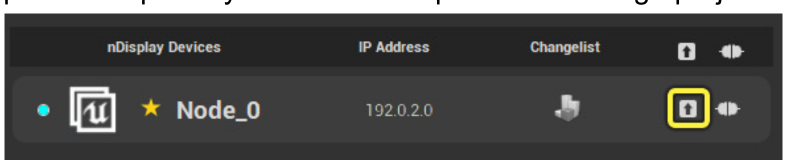
12. Close the **Settings** window.
13. In the Switchboard window, make sure **Level** is set to the correct “game” level of your app.



14. Click the **Connect to listener** button to connect to the **Switchboard Listener.** You may be asked to provide the password that you chose earlier.



15. Start the **nDisplay instance**. This will immediately minimize all windows and launch the nDisplay instance showing the scene. Note: it will take some time for the scene to finally appear, so be patient- especially on slower computers or for large projects.



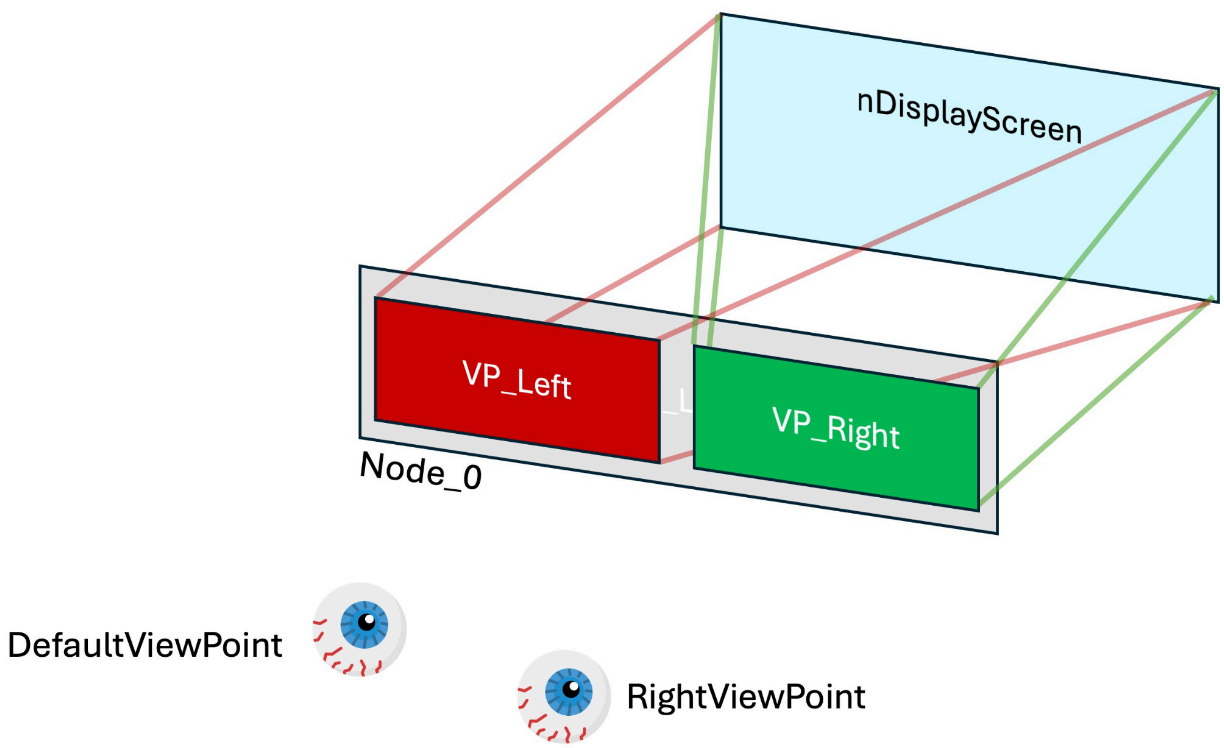
16. Once launched, you should be able to interact with your program. For example, if you started with the nDisplay template project you should be able to use First-Person-Shooter controls (WASD + mouse) to move about the environment.
17. Press the ESC key to exit the application.

Final note: When you use your Switchboard again at a later date, you can quickly bring up the appropriate configuration for your project by clicking on the **Config** menu and selecting the previously created configuration. Don‘t just assume that the configuration you see loaded by default is the correct one, as it may be the configuration for someone else‘s project since your display wall is used by multiple people for multiple projects.

Part E. Set up nDisplay Actor from Scratch for Stereoscopic Rendering on LAVA's Display Wall

This part of the tutorial is optional to most users. It explains how to build LAVA's nDisplay actor from Part C from scratch. However you may also want to read this if you intend to modify the configuration in some way.

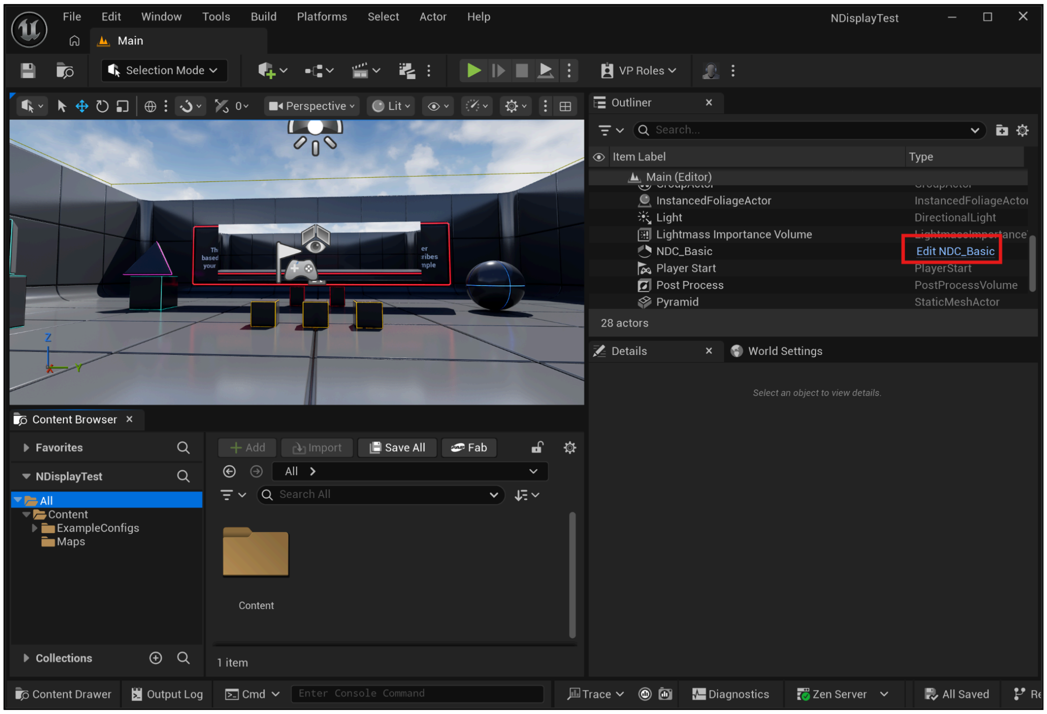
Fundamentally what LAVA's nDisplay asset does is, for each eye of your stereo image, assign each **view point** to a viewport, and the **viewports** map to an **nDisplay Screen**. Whereas the **viewport** specifies the pixel resolution of the wall, the **nDisplay screen** specifies the physical dimensions of the wall.



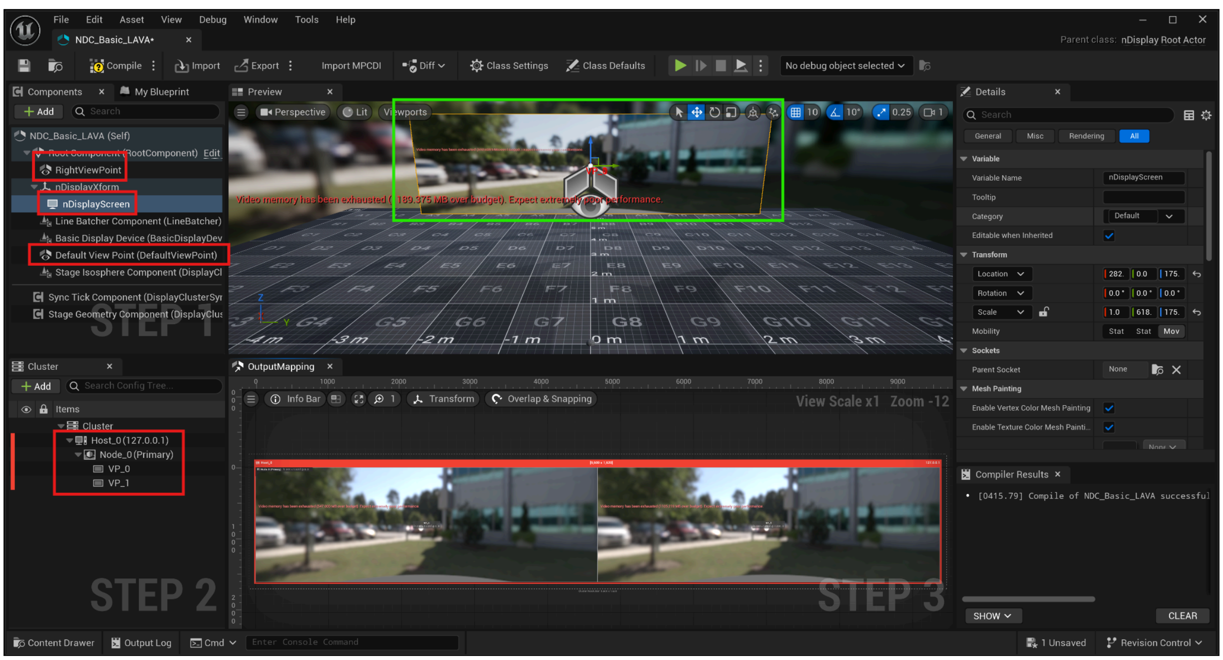
A view point is essentially the location of the user relative to the display wall.
A view port is the region into which the graphics will be rendered.
The nDisplay screen is the virtual representation of the physical display wall.

The steps needed to do this are as follows:

1. Continuing from Part A, edit your nDisplay actor (NDC_Basic) in the Outliner tab of your nDisplay project.



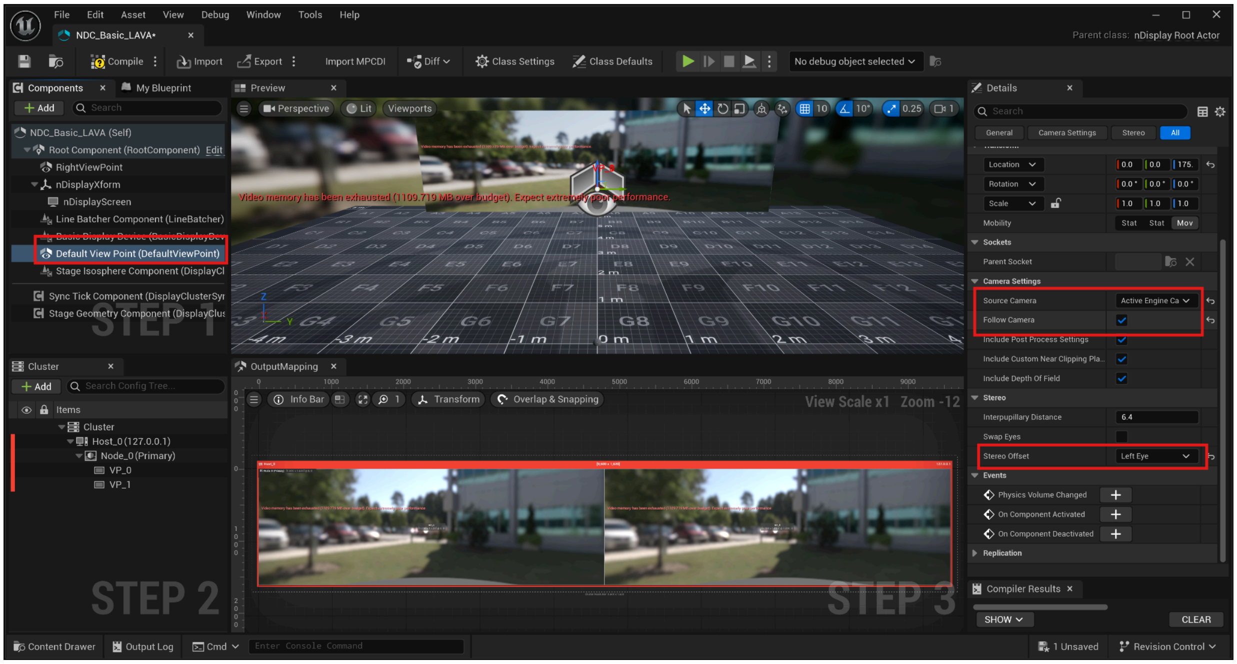
2. This is what the NDC_Basic's editor interface window look like:



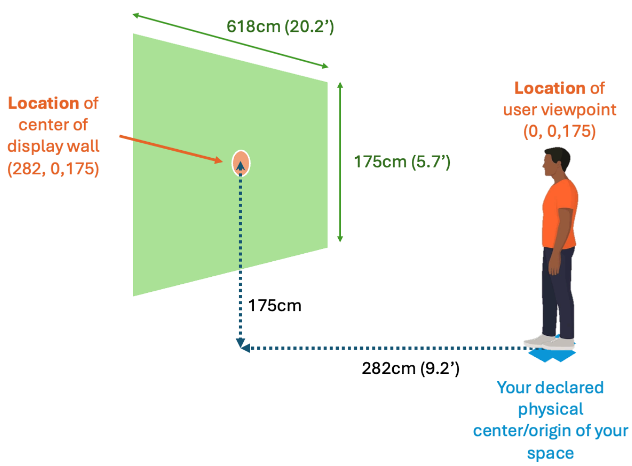
Marked in RED on the left are the nodes we will be modifying. The top left tab shows the components of the nDisplay actor. The bottom left tab shows the nDisplay cluster settings. The Preview window in the center-top shows in green the nDisplay screen. The bottom Output Mapping window shows how the viewports map to Node_0's frame buffer.

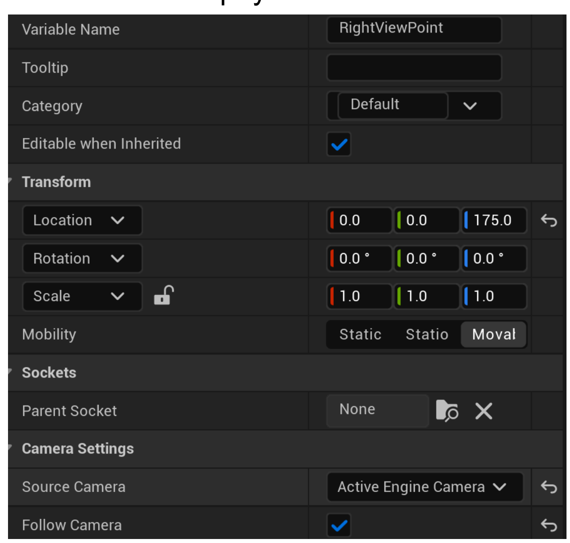
3. We're going to use the **Default Viewpoint** for the **left** eye. So click on Default Viewpoint in the Components panel and in the details panel on the right look for Stereo and set the Stereo Offset to Left Eye.

4. Make sure for the viewpoint, the **camera setting** → **source camera** is set to **Active Engine Camera**, and also make sure **Follow Camera** is checked.

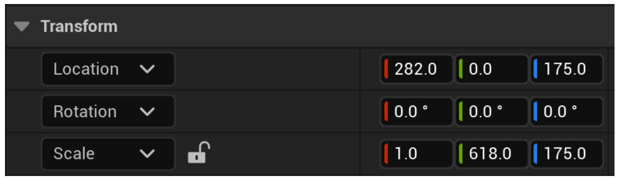


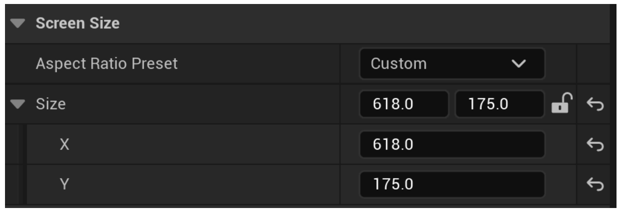
5. We'll create a new viewpoint (and call it **RightViewPoint**) for the **right** eye, by duplicating the **Default Viewpoint** and setting stereo to **right eye**. To duplicate the Default Viewpoint, right click on it to bring up a menu and select **Duplicate**.
6. Set the **Default ViewPoint** and **RightViewPoint**'s location to **0, 0, 175** (5'7" ~ 175cm). This also assumes the adult is standing at the center of their physical world.



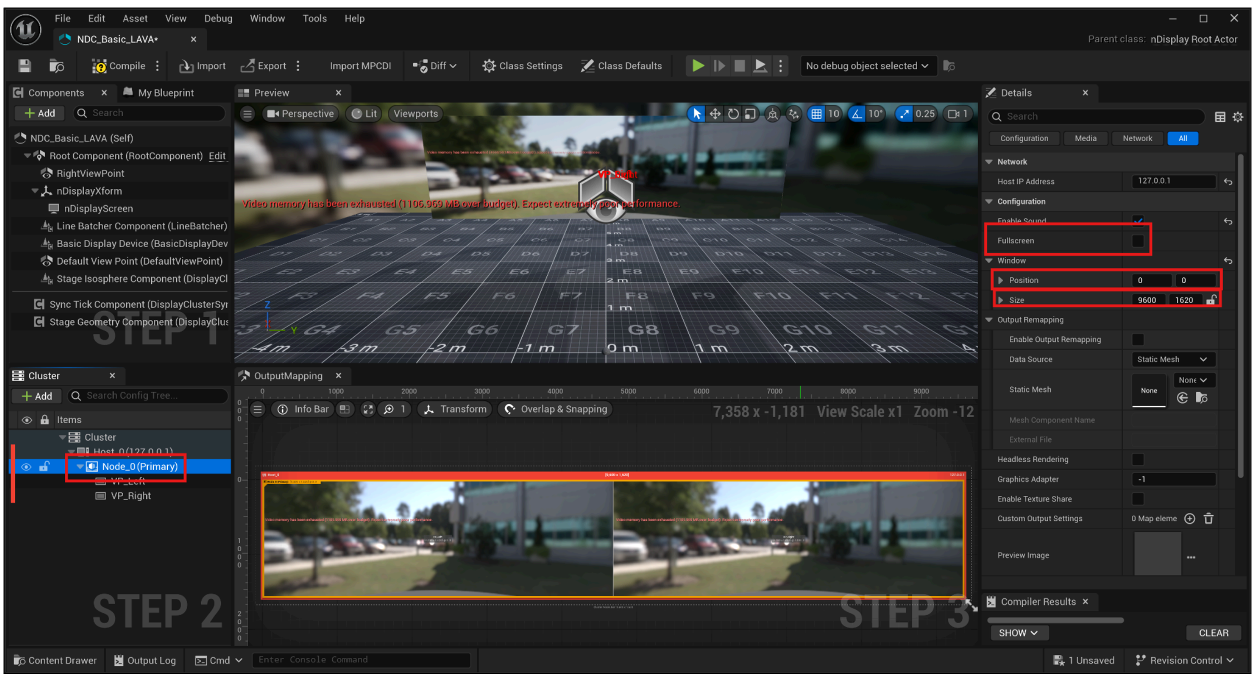


7. Rename VP_0 to VP_Left by right clicking on VP_0 and selecting Rename.
8. Create a second viewport by duplicating VP_Left (call it VP_Right).
9. Set **nDisplayScreen's Transform** and **Screen size** to the values shown below:



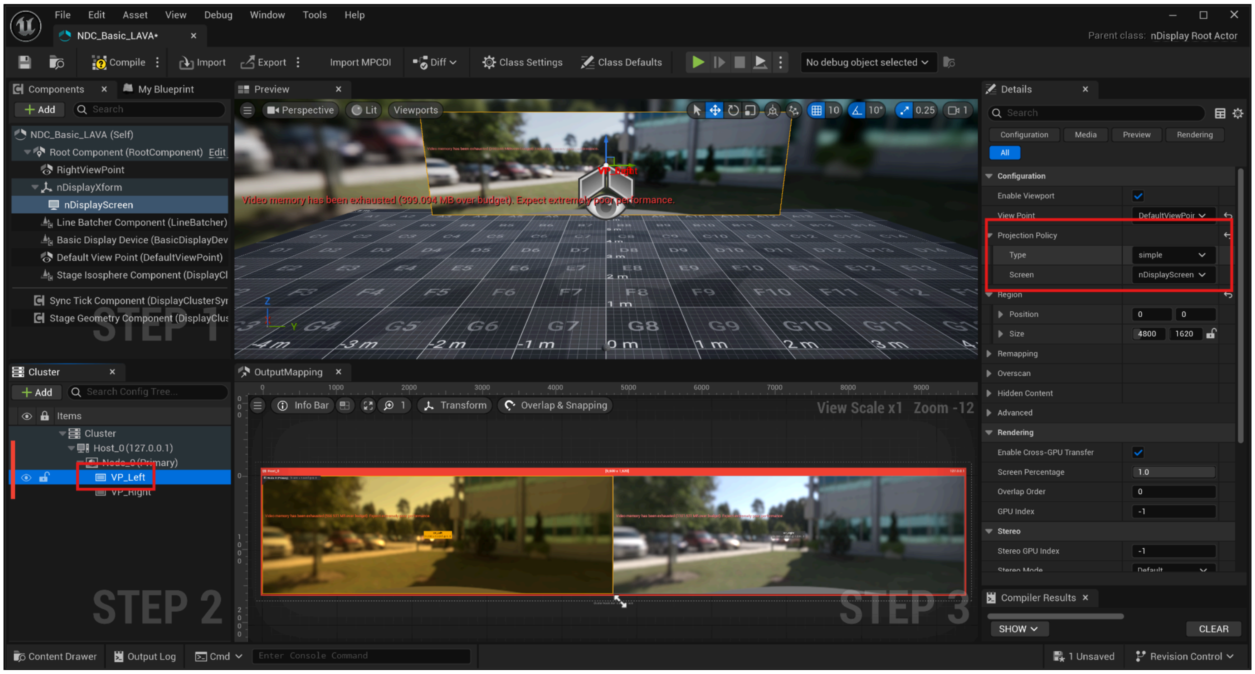


10. This sets the physical screen size to 618cm x 175cm (which is 20.2’ x 5.7’). The screen is also set to 282cm (9.2’) in front of the viewer.
11. For **Node_0** rendering size, set it to the full stereo width (in pixels) of the display wall and the individual viewports (VP_Left and VP_Right) to half the size:
    - For Node_0‘s **Full Screen** setting, make sure it is NOT checked
    - For Node_0 **Size** is 9600x1620

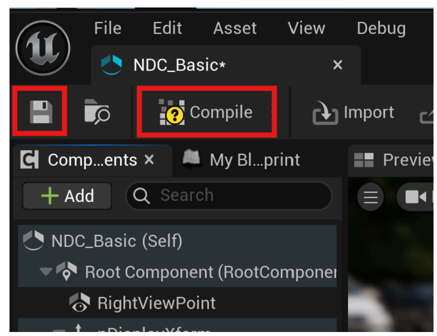


    - For VP_Left **Size** is 4800 x 1620
    - For VP_Right **Size** is 4800x1620
    - For VP_Right **Position** set it to 4800x0

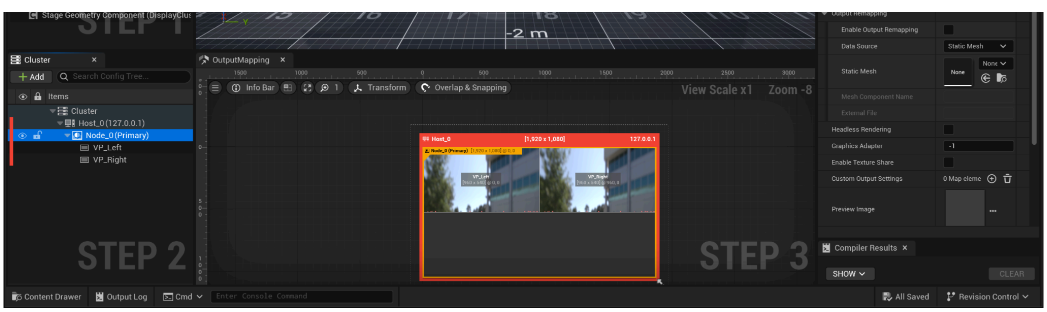
12. In **VP_Left** set the **Configuration/View Point** to **DefaultViewPoint**
13. Set its **Projection Policy/Type** to **Simple** and **Projection Policy/Screen** to **nDisplayScreen**



14. Similarly, for **VP_Right** set the **Configuration/View Point** to **RightViewPoint** and set the **Projection policy** to **Simple**, and **Projection Policy/Screen** to **nDisplayScreen.**
15. After making all the edits, click on Save and Compile in the top left of the editor window. Then close the editor window.



If you're wondering what the settings for the NDC_Sim_LAVA are, Node_0 is set to 1920x1080, VP_Left and VP_Right are set to half those values, like below:



Part F. Tips and Caveats

Running a Packaged App with nDisplay

The above tutorial assumes you will run your nDisplay application from your Unreal editor. However it is possible to build the application to a package and run the executable from Switchboard. All you need to do is to build your game package as you normally would any other game you develop for Unreal, and in Switchboard click on the Settings menu, go to nDisplay Settings and browse and select your executable program.